

Comparing Logistic Regression and Random Forest for Binary Classification

1 Introduction

This study compares a regularized linear classifier with a nonlinear tree ensemble on the supplied tabular dataset. The objective is a reproducible workflow with justified cleaning, preprocessing and evaluation decisions. The positive class is "yes". Model settings are selected using training-only cross-validation; the supplied test partition is reserved for the final comparison.

2 Data exploration and cleaning

The files contain 31,112 training and 13,334 test records, with 15 predictors: seven numerical and eight categorical. Three training records and one test record have missing labels; these are excluded because supervised targets cannot be inferred reliably. The remaining training set contains 31,109 records (7,464 yes; 23,645 no), and the evaluation set contains 13,333 records (3,168 yes; 10,165 no). The positive proportions are 24.0% and 23.8%, respectively, indicating moderate class imbalance.

The original training file has 47 missing predictor cells, fewer than 0.02% of predictor entries. Numerical gaps are filled with training medians, which are less sensitive to extreme values than means; categorical gaps use training modes, a simple choice given the very small missing fraction. No exact duplicate rows occur within either file; 0 test feature rows exactly match training feature rows. These checks cannot rule out repeated entities because no entity identifier is supplied.

Using training-set 1.5-IQR fences, activity_duration has 5,347 flagged values and load_ratio has 2,526. These are statistical tail observations, not verified data errors. They are retained because no data dictionary establishes invalid ranges and deleting them could remove meaningful minority patterns. This conservative choice also leaves logistic regression potentially sensitive to extremes; clipping would require a separate training-only sensitivity study.

3 Features and experimental procedure

All 15 predictors are retained: their names alone do not justify discarding them, and the feature count is modest. One-hot encoding represents categorical values without imposing artificial order; unseen categories are ignored at transformation time. Numerical features are standardized for logistic regression so regularization is comparable across scales; the forest uses unscaled numbers. No interactions, resampling or label-derived features are added. Imputation, encoding and scaling sit inside each model pipeline and are refitted within every validation fold, preventing preprocessing leakage.

Both models use the same shuffled, stratified three-fold cross-validation (seed 42). Grid search selects mean average precision (AP), emphasizing positive-class ranking under imbalance. AP is the recall-weighted average of precision across score thresholds, not trapezoidal PR-AUC. Each selected pipeline is refitted on all labelled training data, then evaluated once on the test set. A fixed 0.50 threshold produces class predictions; no test-based threshold adjustment is performed.

Logistic regression [1] uses L2 regularization, the LBFGS solver, C in {0.1, 1, 10}, tolerance 0.0001 and a maximum of 2,000 iterations. Random forest [2] uses 200 bootstrap trees, Gini splits, sqrt feature subsampling, maximum depth in {12, unrestricted}, and minimum leaf size in {2, 10}. Trees stop at the selected depth or when no admissible split remains; training stops after 200 trees. Both models use default unweighted classes. This preserves observed class proportions but may sacrifice positive recall; no unequal error costs were supplied.

4 Results

The selected logistic regression uses $C = 10$; the selected forest uses maximum depth = 12 and minimum leaf size = 2. Their cross-validation AP values are 0.7085 (fold SD 0.0108) and 0.7130 (fold SD 0.0145), respectively. These fold standard deviations describe split variability, not confidence intervals.

Table 1. Held-out test performance; precision, recall and F1 refer to the yes class.

Metric	Always no	Logistic regression	Random forest
Accuracy	0.7624	0.8396	0.8390
Balanced accuracy	0.5000	0.7473	0.7311
Precision	0.0000	0.6986	0.7211
Recall	0.0000	0.5713	0.5256
F1	0.0000	0.6286	0.6080
ROC-AUC	0.5000	0.8888	0.8895
Average precision	0.2376	0.7025	0.7067

The always-no baseline has zero positive recall and F1 despite 76.2% accuracy, illustrating why accuracy alone is insufficient. Its precision is set to zero by convention because it predicts no positives. Balanced accuracy averages sensitivity and specificity; ROC-AUC and AP assess ranking without fixing a classification threshold. The baseline AP of 0.2376 equals test positive prevalence.

Table 2. Test confusion counts at threshold 0.50.

Model	True no	False yes	Missed yes	True yes
Logistic regression	9384	781	1358	1810
Random forest	9521	644	1503	1665

5 Discussion

On this test split, random forest achieves higher AP. The forest changes AP by +0.0043 and F1 by -0.0206 relative to logistic regression. The comparison tests whether nonlinear splits and interactions add value beyond an additive linear score; these observed differences do not by themselves establish statistical significance or explain which predictors cause the outcome.

At threshold 0.50, logistic regression misses 1,358 positives and produces 781 false positives; the forest misses 1,503 and produces 644, respectively. The practical choice therefore depends on error costs. If missed positives are expensive, a lower threshold or class weighting should be evaluated using training-only validation, with the resulting precision cost reported. Probability calibration has not been assessed, so scores should not automatically be treated as reliable individual risk estimates.

The limited grid and one supplied split constrain generalization claims. The data lack a domain description, entity identifiers and time information; random stratification cannot verify independence or exclude semantic target leakage. No causal interpretation is warranted. A useful next step is repeated or grouped validation if suitable identifiers become available, followed by validation-only threshold selection and a fresh external test. Retained outliers and mode imputation also merit sensitivity checks before deployment.

6 Conclusion and reproducibility

The workflow demonstrates that minority-sensitive metrics reveal performance hidden by the majority baseline. Random forest provides the stronger observed AP, while the confusion counts make the operational trade-off explicit. Results are reproducible with seed 42 and scikit-learn 1.9.1; the accompanying script exports every grid result, prediction and audit count.

Source repository: <https://github.com/Davi41234/IN6227>

References

- [1] Scikit-learn. LogisticRegression API documentation. [Official documentation](#).
- [2] Scikit-learn. RandomForestClassifier API documentation. [Official documentation](#).